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Response to Comment on “Posterior comparison of model dynamics in several hybrid turbulence model forms” [Phys. Fluids 38, 069101 (2026)] **FREE**

Colin AZ Towery ; Juan A Sáenz ; Daniel Livescu  *Physics of Fluids* 38, 069102 (2026)<https://doi.org/10.1063/5.0324470>

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In our recent work, Towery *et al.*,¹ (hereafter referred to as TSL24), we developed a novel verification and validation (V&V) test framework for assessing hybrid turbulence models in a non-wall-bounded and non-stationary flow context using decaying homogeneous isotropic turbulence (HIT). Models tested combine the two-equation Reynolds-Averaged Navier–Stokes (RANS) model form (e.g., k - ε) with the scale-resolving capabilities of large-eddy simulation (LES) by modifying one or more RANS model terms with a resolution-sensitive coefficient. We applied this V&V test to a wide-array of such hybrid RANS-LES (HRL) models, including the Flow Simulation Methodology (FSM), Detached Eddy Simulation (DES), Partially averaged Navier–Stokes (PANS), Partially Integrated Turbulence Model (PITM), and Continuous Eddy Simulation (CES). Our conclusions were that FSM-like models were always dynamically inconsistent with the direct numerical simulation (DNS) ground truth; that DES, PANS, CES, and other model forms similar to these all exhibit the same qualitative dynamical behavior in simulating decaying HIT; and that no model was universally superior across all grid resolutions.

In a Comment,² the author raises the point of how the CES model was used in TSL24 and mentions the results obtained in wall-bounded flows,^{3,4} where the model was shown to outperform other hybrid models. We thank the author of the Comment² for giving us the opportunity to better explain our procedure and the performance of the analyzed models.

The arguments in Comment² are summarized as follows: (i) that in approximating the modeled kinetic-energy fraction F_k using a von Karman spectrum relation, $R_{vK} \propto (\mathcal{L}_t/\Delta)^{-2/3}$, we have contradicted a defining property of the CES framework, (ii) that we unfairly assumed $F_\varepsilon \approx 1$ in CES but not PANS, (iii) that other studies of CES using complex wall-bounded flows in Fagbade and Heinz^{3,4} (hereafter collectively referred to as FH24), which show CES to perform more accurately than PANS and DES, supersede our results, and (iv) that

therefore our description of CES as being both PANS-like and DES-like is misleading to readers.

In this Response, we address claims (i) and (ii) by showing that the use of the $F_L^2 = F_k^3/F_\varepsilon^2 \approx R_{vK}^3$ approximation is not a significant source of error for our test case, and therefore does not lead to any additional “actual-imposed resolution mismatch” in the CES model tests compared. We further address claims (iii) and (iv) in a discussion of the important differences between the physical configurations and analyses of TSL24 and FH24, indicating that these three studies are fully complementary and consistent with each other and that a careful reader is not being misled by our results.

As a starting point for this response, we recall our definition of the generalized k - ε transport equations, Eq. (10) of TSL24,¹ wherein one or more of the RANS model constant coefficients is replaced with a resolution-sensitive, non-dimensional function (denoted as C^*), and the coefficient C_{k2}^* (named by analogy to $C_{\varepsilon2}$) was added to Eq. (10a) in comparison to the standard RANS. Individual models were categorized *a priori* by the RANS coefficient they modify: FSM-like if modifying only the C_μ coefficient, DES-like if utilizing a non-unity C_{k2}^* (alongside any other modifications), and PANS-like if modifying only the $C_{\varepsilon2}$ coefficient. These categorizations did not depend on *posterior* model performance or feedback mechanisms. More specifically, all PANS-like models use a modification of the form $C_{\varepsilon2}^* = C_{\varepsilon1} + R(C_{\varepsilon2} - C_{\varepsilon1})$, with a resolution indicator, R .

In the original PANS model derivation,⁵ $R_{PANS} = F_k/F_\varepsilon$, where F_k and F_ε are the observed fractions, just as in CES. Similarly, the original PITM derivation⁶ finds $R_{PITM} = F_k$. The choices by the original authors in each case to substitute approximations to R are argued on practical grounds, not theoretical. In the case of PITM, the approximation R_{vK} is sensitive to $\mathcal{L}_t$, which is an observed value, computed on-the-fly, so that the model resolution R is not arbitrarily imposed on the simulation.

Our choice to use the approximation $F_L^2 \approx R_{vK}^3$ for CES relied first on the fact that $F_k < F_\varepsilon$ is always true when Δ is coarser than

DNS, and therefore $F_k^3 \ll F_e^2$, and secondarily on early test runs of our simulation code indicating that substituting R_{VK}^3 for F_L^2 had little effect on the results. In addition, we were interested in directly comparing CES-K and CES-X to DES and XLES, respectively, where all models had an explicit dependence on Δ . We regret that we did not explain these choices sufficiently clearly in our publication, and the preceding comment is helpful to point out this oversight.

Figure 1 compares new test data for CES-S- F_L^2 (i.e., using the observed F_L^2 as R) and PITM- F_k to the previously published data for CES-S- R_{VK}^3 , PITM- R_{VK}^3 , and PANS- F_k/F_e (note again that both F_k and F_e are observed outputs in the PANS and PITM cases). It can be clearly seen that the two CES-S cases are virtually identical at filter wavenumbers $\kappa_f = 16$, and 128, and only slightly different at $\kappa_f = 2$. The initial value for F_e at each κ_f is $F_{e,0} = 1.0, 0.97$, and 0.36 , respectively. We further provide and discuss quantitative error metrics for F_k and F_e at all κ_f in the [supplemental material](#).

A main goal of TSL24 was to provide a V&V testing framework for HRL models in a non-wall-bounded flow and with a very accurate numerical method on a uniform grid to eliminate any effects of near-wall model interventions, numerical discretization errors, and mesh geometry dependence. As such, we were able to test HRL models across the full range of $0 < F_e \leq 1$. Furthermore, we tested model feedback mechanisms in a non-stationary flow where, for a fixed Δ , the true resolution indicators (F_k , F_e , F_L) are constantly changing. Finally, the emphasis in our discussion and conclusions regarding the similarities between DES, PANS, PITM, and CES were clearly stated to be based on qualitative agreement in the coefficients they modify, and the feedback mechanisms they induce, not equivalent accuracy levels. Where deficits in CES were pointed out, they were in the modeled quantities $\mathcal{K}_m$ and $\mathcal{E}_m$ when $F_e < 1$.

In contrast, the FH24 V&V studies utilized statistically stationary wall-bounded flows at high free-stream Reynolds numbers. As

such, HRL models were free to respond to both the inflow turbulence and wall boundary layer over very long times until they reached a statistical-steady-state condition in their computed R . Moreover, all test cases shown exhibit $F_e \approx 1$ except within the tiny region of the inner boundary layer. Finally, FH24 focused on the quantitative accuracy in matching to DNS results of mean-flow data, such as the location of flow separation and reattachment and the mean velocity profiles.

The Comment and FH24 argue that by directly using resolution indicators in model coefficients, the CES models suffer no “imposed-actual resolution mismatch.” While it is true, by definition, that any HRL which dynamically adapts its resolution control parameter R to match the actual solution will have no “imposed-actual” mismatch, that is no guarantee that the HRL solution will match the “true” solution of a comparable DNS. When a flow configuration is statistically stationary like those in FH24, it may be possible for the HRL to evolve over time to reach an equilibrium wherein the actual resolution matches the true DNS resolution. In contrast, if an HRL model imposes some approximation to the target resolution, there is no guarantee the actual resolution will ever evolve to match either the model-imposed target or true resolution. In these flow configurations, “imposed-actual resolution mismatch” is an important criterion for evaluating model self-consistency.

In the case of a flow configuration where there is no dynamical equilibrium, such as decaying HIT, TSL24 and Fig. 1 clearly demonstrate that there can be a large mismatch between HRL and DNS resolution state-space trajectories (i.e., an “actual-true resolution mismatch”), even when there is no “imposed-actual” mismatch by construction. As such, “imposed-actual resolution mismatch” is no longer a sufficient criterion by which to judge model fitness. To further illustrate this point, we present time series for the “imposed-actual,” “actual-true,” and “imposed-true” resolution mismatch ratios in the [supplemental material](#).

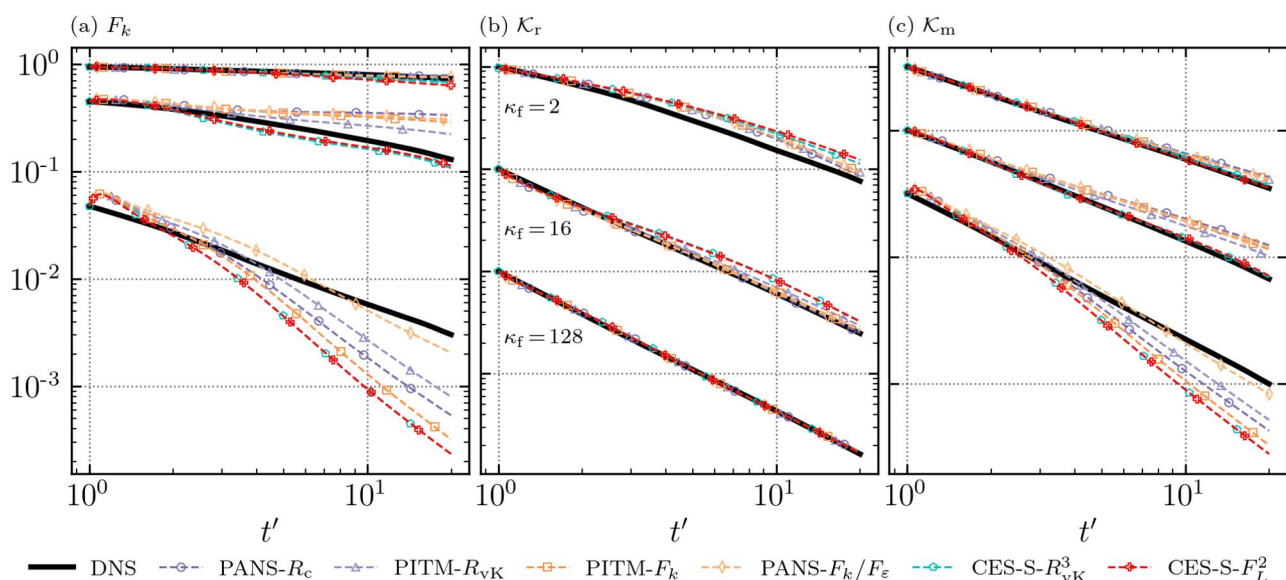


FIG. 1. Time series of (a) observed F_k output, (b) resolved kinetic energy, and (c) modeled kinetic energy for filter wavenumbers $\kappa_f = 2, 16$, and 128 (corresponding to almost-URANS, mid-range LES, and almost-DNS). Panels (b) and (c) include an arbitrary offset along the y-axis for each κ_f value to separate series visually.

In summary, the differing flow configurations between FH24 and TSL24 make for complementary tests to each other, as do the analytical frameworks. When simulations can appropriately evolve to statistical equilibrium, resolution indicators like F_k are sufficient quantities of interest on their own. However, when no stationarity exists, we find it necessary to analyze the individual dynamical terms in the underlying differential equations for $\mathcal{K}_m$ and $\mathcal{E}_m$ that govern the resolution indicators to fully understand model behavior. Therefore, TSL24 and FH24 do not conflict in results or conclusions, and TSL24 does not mislead the reader.

SUPPLEMENTARY MATERIAL

See the [supplementary material](#) for error metrics for F_k and F_ϵ at all κ_f along with time series for the “imposed-actual,” “actual-true,” and “imposed-true” resolution mismatch ratios.

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Office of Experimental Sciences (OES), as well as computational resources from LANL’s Institutional Computing (IC) program.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request and the DNS code is openly available in GitHub at <https://github.com/lanl/ShenCFD>, Ref. 7.

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